

MAE Lab 1 Report: 3D Printed Glider

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This report details the design, development, and testing process for a 3D-printed aircraft created as part of the MAE 339 Aerospace Engineering Laboratory at the University at Buffalo. The project tasked students with designing an aerodynamic, aircraft-like model using only 3D-printed components, glue, and paint, with the goal of achieving maximum glide distance while maintaining structural integrity. Design choices, material selection, airfoil shape, and structural infill percentages, were meticulously justified based on aerodynamic principles and manufacturing constraints. The project culminates in a competition where models are evaluated based on their flight performance and durability. This paper explores the methodologies employed, results obtained, and lessons learned during this iterative design process.

I. Nomenclature

AOA	=	Angle of Attack
C_d	=	Coefficient of Drag
C_{HT}	=	Horizontal Tail Volume Coefficient
C.G.	=	Center of Gravity
C_l	=	Coefficient of Lift
$C_{m,c/4}$	=	Pitching Moment Coefficient
C_{mgc}	=	Mean Geometric Chord
C_p	=	pressure coefficient
C_{VT}	=	Vertical Tail Volume Coefficient
D	=	Drag
h	=	Height
L	=	Lift
S	=	Wing Area
V	=	Velocity
W	=	Weight
ρ	=	Air Density
μ	=	Dynamic Viscosity of Air
γ	=	Flight Path Angle

II. Introduction

This project focuses on the design, fabrication, and testing of a 3D-printed glider as part of the MAE 339 Aerospace Engineering Laboratory at the University at Buffalo. The goal was to create an aircraft-like model capable of achieving the maximum glide distance during a competition while maintaining structural integrity. The design had to adhere to specific guidelines, including using only 3D-printed materials, glue, and paint. The competition involved

launching the gliders, with the winner determined by the farthest distance covered and survival after landing. All the gliders were meant to be launched by hand from the same starting point and at the same time. The glider had to be carried by lift rather than throwing power, and only the final flight distance determined the ranking, provided the model survived landing. Our glider featured a high-wing design for enhanced stability, a cruciform tail for effective control, and an elongated, conical nose to minimize drag and improve aerodynamic performance. Lightweight, thin-walled 3D-printed components were used to reduce weight and optimize glide performance, while the entire aircraft was reinforced with epoxy paint for added strength. Despite meticulous planning and testing, our glider performed well in flight but ultimately crashed on landing, highlighting both the strengths and areas for improvement in our design. This report provides a detailed account of the design methodology, testing outcomes, and engineering principles applied throughout the project, along with reflections on the lessons learned.

Each team member contributed significantly to the project, bringing diverse skills and expertise to the design process:

Sean spearheaded the analytical aspects of the project, conducting critical calculations to ensure the gliders design met the aerodynamics and structural requirements. He also created the SolidWorks model, translating the team's conceptual design into a detailed 3D representation, ensuring it adhered to the competition guidelines.

Raisha played an important role in the assembly of the glider, working meticulously to combine the printed parts into a functional and structurally sound model. She also played a pivotal role in collecting flight data during the test flights, ensuring it met the team's performance expectations and guided further improvements.

Humza was integral to the printing process, ensuring the SolidWorks model was successfully translated into a physical component. He oversaw materials selection, printer settings and quality control during production. His attention to detail helped prevent issues during printing resulting in high quality components for the glider.

Frank partnered with Humza in managing the printing process, coordinating printer schedules and troubleshooting technical challenges. He also ensured that the printed components met the team's specifications and were ready for assembly. Humza's contributions ensured a seamless transition from digital design to physical construction.

Vlad played a pivotal role in the aerodynamic analysis, optimizing the gliders airfoil shape and overall design for maximum glide performance. He also contributed to determining the optimal infill percentage, balancing structural integrity and weight. His efforts were key in refining the design to achieve best possible flight performance.

Ayesha played a key role in the assembly process, focusing on reinforcing critical joints and ensuring structural integrity. She also contributed to pre-flight checks, test preparations and final adjustments, helping optimize the gliders performance and readiness for the competition. Her contributions to assembly and testing were vital to the project.

Despite these specific roles, all team members actively participated in every aspect of the project, including brainstorming, build hours and test flights. The collective effort and consistent attendance of all members ensured a cohesive and collaborative approach to the project.

III. Technology and Literature Review

A thorough technology and literature review of various resources was conducted to establish an informed approach comprised of empirical data, trade studies, analysis tools, and flight tests. Initial considerations were focused on aerodynamic efficiency and the specific strength of structural elements. These considerations were successfully implemented with the support of the following methods and programs. Trade studies and empirical data informed the starting point of the iterative process through which analysis tools were employed to further refine desirable output of design parameters.

The trade study found several examples of relevant aircraft designs from which inspiration for this project was derived. Hand launched gliders designed by Dr. Mark Drela including the supergee and apogee were examined

as well as 3D printed glider developed and sold by Soarkraft. The Drela aircraft established the baseline of airfoil performance for comparison, whereas Soarkraft designs demonstrated the expected output with respect to mass considerations and proportionality. Both sources provided relevant examples regarding scale and informed the initial aircraft sizing, which have wingspans between 1.5 and 2 meters.



Fig. 1: Apogee by Drela

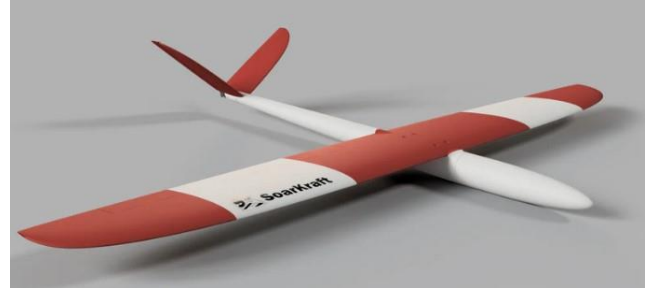


Fig. 2: Pika by Soarkraft

To determine an appropriate mass distribution, a MATLAB script was developed to approximate the weight of discrete sections of the fuselage. Output from this analysis tool was used in conjunction with other modeling software to attain an optimized Lift to Drag ratio (L/D) of the complete airframe.

A potential flow solver, FLOW5, was used to further resolve the initial sizing and expected performance. FLOW5 utilizes the non-linear lifting line theory and a combination of two vortex-lattice methods along with the volume-panel method allowing for preliminary wing and aircraft performance estimation.

The development of individual components was accomplished via SolidWorks CAD software. Methods employed to generate the various components were researched and several video resources were found to inform the production significantly. Fuselage design was executed using methods published by 3D CAD Freak, while wing structures were developed following instructions posted by C. Brindle. The use of a set of threaded prints were made using methods documented by GoEngineer.

The 3D printer used for this project was a Bambu Lab A1. This FDM printer has a build volume of 256mm * 256mm * 256mm and, due to the all-metal hot end, allows for the use of various high temperature printing filaments. For this project a 0.4 mm and 0.6 mm hardened steel nozzle were used to produce all parts.

To prepare CAD output files for the printer, Bambu Studios slicer was used to ensure desired infill percentage, required print speeds and the necessary flowrates for the various materials utilized for this project.

Research on airframe surface finish was found noting that aircraft surfaces are painted with either enamel or epoxy paint. The choice of a two-part epoxy paint, similar to the offering found at Boeing, was chosen due to the finish quality, durability, and its compatibility with 3D printed materials.

IV. Conceptual Design

The conceptual design of the 3D-printed glider focused on achieving a balance between aerodynamic performance, manufacturability and durability while adhering to the competition guidelines. One of the first steps before selecting different configurations for an aircraft is to first visualize how a typical glider works using a free body diagram.

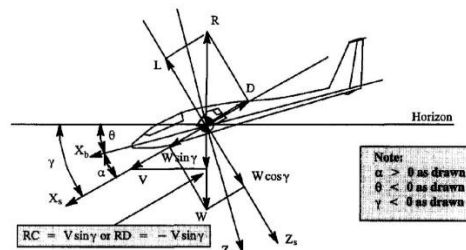


Fig. 3 Forces Acting on an Airplane in Un-powered Flight

The figure shown above depicts a glider freely gliding along a flight path γ . While there are only 3 fundamental forces acting on the glider (Lift, Drag, Weight), the flight path angle causes the Lift and Drag Vectors to be adjusted as well to the flight path angle. Knowing that the flight path angle is a relationship of height and glide distance such that:

$$\tan(\gamma) = \frac{h}{d} \quad (1)$$

Relating Lift and Drag Forces to the components of the Weight force allows the Equation 1 to be simplified so that:

$$d = h \frac{L}{D} \quad (2)$$

$$d_{max} = h \left(\frac{C_L}{C_D} \right)_{max} \quad (3)$$

This shows that weight isn't a dependent factor in maximizing distance at all. Only the aerodynamic performances of the aircraft affect the overall glide performance. Considering the aircraft takes flight in sub sonic speed, the relationship between C_L and C_D are only affected by the AOA. Additionally, the optimal glide speed needed for the glider to perform for its maximum distance can be solved by using Lift, Drag, and Weight as a Pythagorean relationship:

$$W^2 = L^2 + D^2 \quad (4)$$

Knowing that Lift and Drag can be turned into non-dimensional coefficients, the equation can be rearranged into

$$V = \sqrt{\frac{2W}{\rho S \sqrt{C_L^2 + C_D^2}}} \quad (5)$$

Overall, with Equation 2 and 5, the team has discovered that the only maximizing L/D is the only main factor that should be considered in selecting a configuration in which velocity for that specific L/D may be calculated. Therefore, a conventional configuration was selected for its well-documented stability and predictable flight characteristics, simplifying the optimization process for glide performance. The high wing design was chosen to enhance roll stability by lowering the center of gravity relative to the center of lift, which promotes smooth, level flight and minimizes risk of uncontrolled rolls or spins. Additionally, the high wing design allowed the glider to be held underneath the fuselage, providing a better grip for improved throwing accuracy and consistency. The cruciform tail configuration provided effective pitch and yaw control while avoiding airflow interference from the main wing. The design of the glider incorporated a 4-degree dihedral angle in the wing configuration, enhancing self-correcting flight characteristics. This arrangement ensured efficient control surface performance and reduced the risk of damage during landing or uneven terrain. The fuselage had a tadpole shape to minimize aerodynamic drag and wake turbulence, ensuring smoother airflow transition and reduced pressure drag. The streamlined fuselage was complemented by hollow, thin-walled 3D-printed parts to reduce weight and material usage staying within the project printing constraints. The 3D-printed aircraft components were attached together with extra thick plastic adhesive glue to ensure proper bonding and improve structural integrity and durability.

V. Preliminary Design

A. Airfoil Selection

Preliminary glider design began with airfoil trade studies for the wing and tail. After reviewing the available literature on low Re airfoils suitable for glider wings, the RG15, Eppler 387, and Selig S7012 were compiled for direct

foil analysis in XFLR5. The results are given by figures X through X, where the operational Re limits were computed using equation 6 to be approximately 50000 and 100000 given extremities of 75 and 180 mm for the mean geometric chord length at the target velocity of 8.94 m/s.

$$Re = \frac{rVL}{\mu} \quad (6)$$

The main objective of the analysis was observing the C_l/C_d profile of each airfoil, and the lift and pitching moment coefficient characteristics are primarily included for completeness as neither the maximum lift nor tail sizing are critically inhibiting parameters. Although the E387 attained the highest C_l/C_d of 62 at a Re of 100000, the RG15 and S7012 were favored because they have a greater average C_l/C_d and operate optimally at lower angles of attack. Despite the similarities between the S7012 and RG15, the S7012 was ultimately selected because a future analysis of 3D aircraft attained a greater L/D using that airfoil.

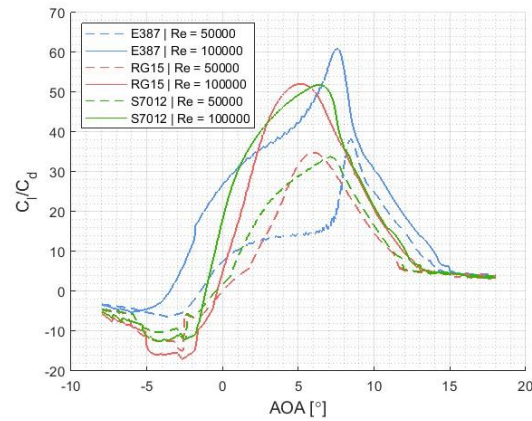


Fig. 4 C_l/C_d vs AOA for E387, RG15, and S7012 airfoils at Re = 50000 and Re = 100000

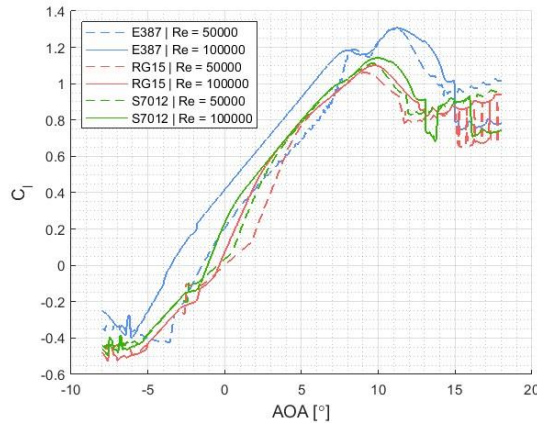


Fig. 5 C_l vs AOA for E387, RG15, and S7012 airfoils at Re = 50000 and Re = 100000

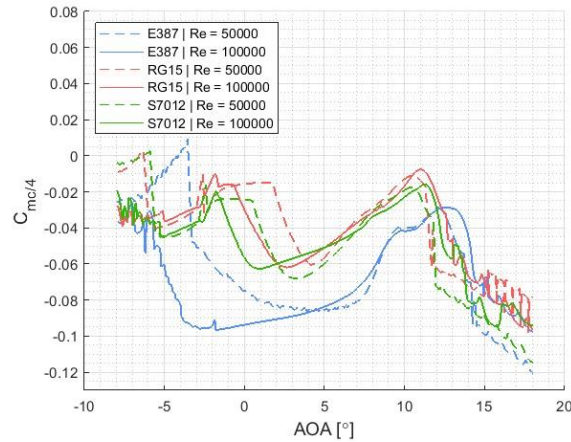


Fig. 6 $C_{mc/4}$ vs AOA for E387, RG15, and S7012 airfoils at $Re = 50000$ and $Re = 100000$

The HT12—a symmetric airfoil with 5% thickness designed by Dr. Mark Drela, the creator of XFOIL, to be used on gliders—was selected for both stabilizers. Symmetrical airfoils are commonly used for tails on aircraft because of the approximately zero pitching moment coefficient at their aerodynamic center; Although assigning negative camber to the horizontal stabilizer would've resulted in a reduction of trim drag, it wasn't deemed worthwhile to pursue and conventional was favored.

B. Aircraft Sizing

After assessing 3D printing methodologies and converging to a design concept, the team wrote an executable MATLAB script to derive the mass and center of mass of a glider model given input dimensions and materials. In accordance with the design methodology, all of the components besides the nose were assumed to exclude infill in favor of outer walls. The walls typically constituted 1 to 2 layers, where the thickness of a single layer is 0.44 mm to closely match the output of a 0.4 mm nozzle.

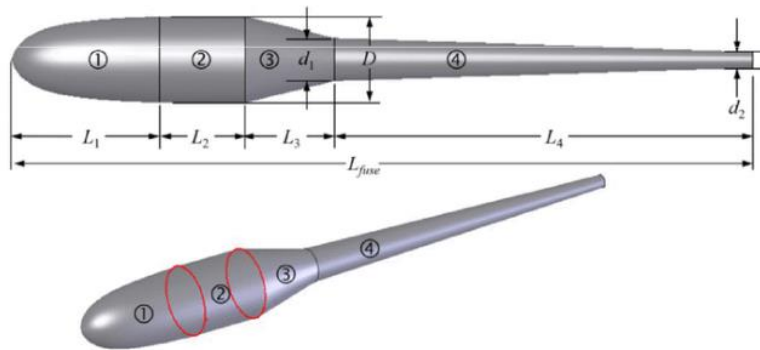


Fig. 7 Tadpole Fuselage Breakdown

The wing and tails were modeled by numerically integrating the individual arclengths of the airfoil splines over the span of the surface and multiplying by an effective linear density that factored in the layer thickness. The tadpole fuselage was modeled using elementary solids depicted in figure X, where section 1 represents a paraboloid, section 2 a cylinder, and sections 3 and 4 are frustums. Other miscellaneous inclusions were a wing spar modeled as a slender rectangular prism, fuselage formers modeled as rings, added nose weight modeled as infill, and an epoxy coat modeled as a percentage of the glider weight. To ensure the feasibility of any given design, nose weight was adjusted as an input to ensure the C.G. was close to the quarter chord of the mean geometric chord of the wing. After attaining a valid design, the parameters from the script were entered into Flow5 to evaluate the L/D and speed polar of the aircraft. From there, the design passed through several MATLAB and Flow5 iterations until achieving a satisfactory sizing.

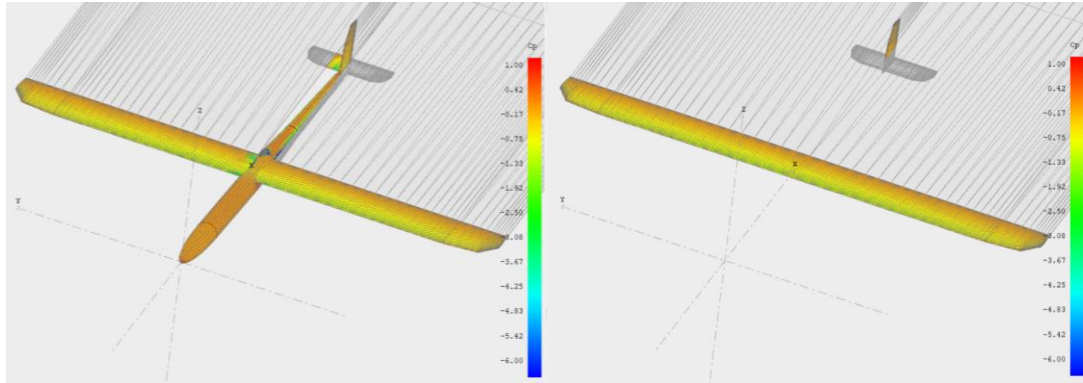


Fig. 8 Cp Distribution and Wake Panels of First Design With and Without Fuselage at 4° AOA

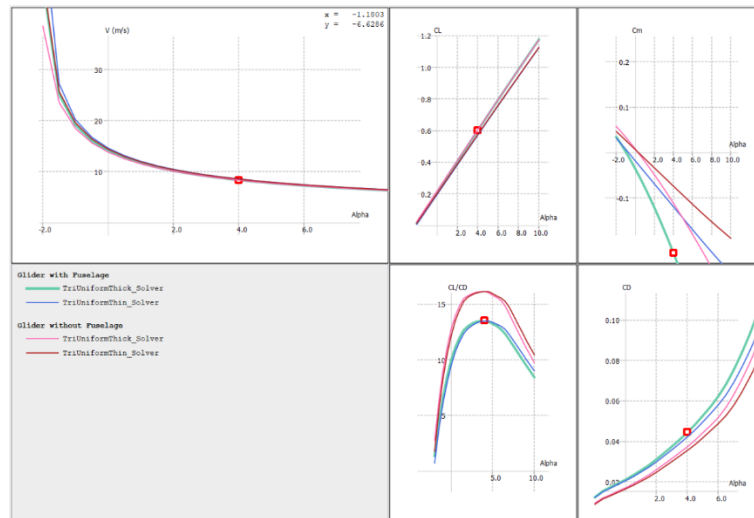


Fig. 9 Aerodynamic Envelope of First Design

The initial Flow5 tests were conducted with a low AR wing because the large disparity between the C_l/C_d peaks at Re values of 50000 and 100000 isn't prevalent on full scale aircraft where Re is typically on the order of 10^6 ; the XFLR5 airfoil analysis gave the impression that having a high AR (effectively a shorter chord) wouldn't be efficient even though induced drag is reduced, yet later analysis indicated that wasn't the case. As such, the first full configuration had a wingspan of 1.180 m, AR of 10.95, projected mass of 340 grams, target L/D of 13.54 at an AOA of 4° and target velocity of 8.41 m/s. The results are summarized by figures X and X, which also serve to validate the analysis by indicating that the solution remained the same regardless of solver type and fuselage inclusion besides a difference in drag. The vertical position of the horizontal stabilizer was lowered from the cruciform position discussed in conceptual design to prevent meshing errors in Flow5, though that wasn't considered to have a considerable effect on the aerodynamics besides the interaction between tail and wing and resultant equilibrium angle. Because the plan was to use a trimmable elevator, the latter point wasn't an important aspect of the analysis regardless.

Subsequent to the first design, the nose was shortened to decrease drag and have a heavier plane with the intent to take a rougher landing from flying faster. As such, the new and final design, shown in figure X along with relevant plots, had the nose shortened by 150 mm and featured an extra 52 grams of nose weight to ensure a sufficient static margin. Because of the increase in weight, the wing incidence angle was set to -2° to ensure that the glider flies close to the target velocity of 8.94 m/s. Observing figure X, that change came with the added benefit of increasing the L/D of the aircraft to 14.34 when flying at a velocity of 8.98 m/s at an AOA of 6°.

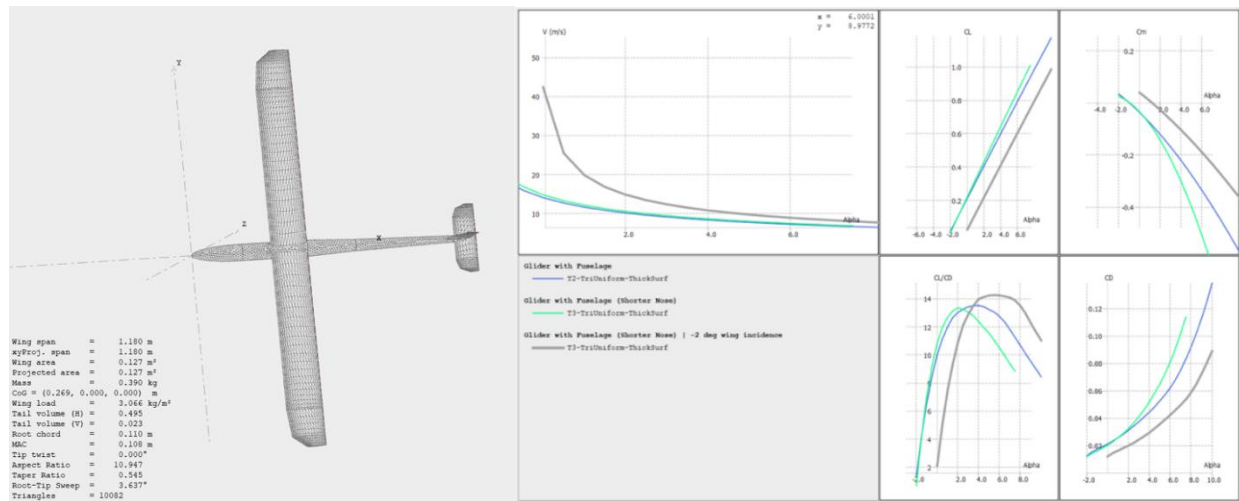


Fig. 10 Second Design Plane (Left) and Aerodynamic Envelope (Right)

The percent mass of each part of the final preliminary design of the glider is given in figure X below, where the total weight of the glider summed to 390 grams. Table X lists out the key specifications for the glider. The vertical tail was sized to be 0.023, which is at the low end for conventional aircraft because directional stability wasn't a massive concern given the objective of gliding; The horizontal tail was initially sized to be 0.495 because of historical guidelines and validated to be reasonable by having the longitudinal static stability derivative be evaluated to -1.4 rad^{-1} , which is within the acceptable range of -0.3 to -1.5 rad^{-1} .

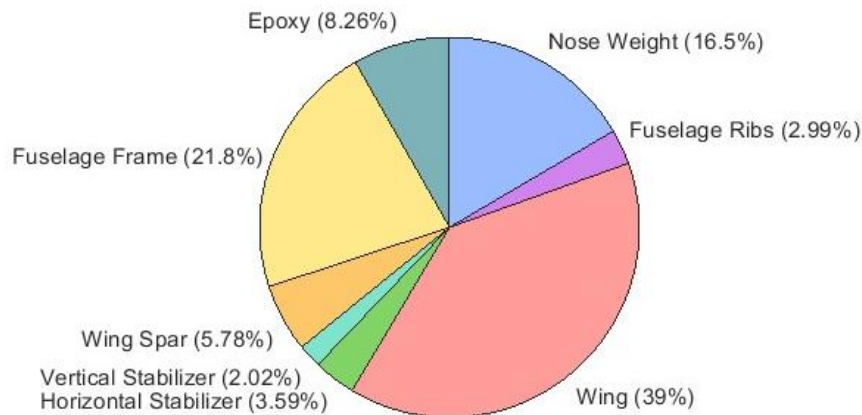


Fig. 11 Component Mass Breakdown

Table 1. Preliminary Design Specifications

Specification	Value
Mass	0.390 Kg
C.G. Location	26.3 % C_{MGC}
Wingspan	1.180 m
Wing Area	0.127 m ²
Aspect Ratio	10.95
Wing Incidence Angle	-2.00°
Wing Dihedral Angle	4.00°
Fuselage Length	0.850 m
C_{HT}	0.495
C_{VT}	0.023
AOA	6.00°
C_L	0.60
C_L/C_D	14.34
Velocity	8.98 m/s
Stall Speed	7.02 m/s

VI. Detailed Design

A logical assessment of the glider's ability to survive a belly landing was conducted using engineering principles and observations from flight tests. One of the project's goals was to design a glider capable of flying steadily and for a sufficient duration to ultimately reach and collide with the designated "wall" target. However, this goal shifted to simply achieving controlled flight long enough to meet this objective. Future projects could benefit from earlier testing phases to address stability and structural issues more effectively, increasing the likelihood of both steady flight and survivability.

A. Materials

To increase the likelihood of survivability after conducting all the flight tests, 3D printed external stringers were glued on the areas that broke during flight tests. The glider components were printed out from the material that complimented their purpose. The material choices and the reasoning behind why particular parts were manufactured from them are as follows:

- 1) **Lightweight PLA (LW-PLA):** One of the main reasons as to why this material was used is its low density and light weight properties. This filament also has a high foaming volume ratio of 220%, which equates it to double the mass of the standard PLA. Although the specific strength of this material is not as desirable as compared to filaments like PLA+, LW-PLA was chosen to print parts. The low specific strength of the material was greatly improved by applying epoxy paint on the part surface.

The parts were mainly printed with no infill, except for the custom "X" pattern infill and ribs in the wing sections, and formers in the hollow tube fuselage sections. All the large bodies like the fuselage sections, wing sections, and the tail sections (including the vertical and horizontal stabilizers) printed using LW-PLA in order to keep the glider as light as possible. This filament costs \$39.14 per kilogram.

- 2) **PLA+:** This filament was used because of its flexibility and toughness characteristics. These properties were necessary for key structural components like the main spar for wing, internal and external stringers for the fuselage sections. PLA+ is substantially stronger than the standard PLA. The filament can withstand more bending stress before it breaks as compared to standard PLA, while having roughly the same density. As a result, components were printed using 100% infill to ensure strength and survivability of the glider. However, the front most fuselage section (where the nose is screwed in) was also printed with this filament to keep the C.G. in a desirable position while also adding strength. Total cost was \$15.99 per kilogram.

- 3) **Brass filled HTPLA:** Since the nosecone was positioned somewhat closer to the C.G. of the glider, a heavier material was needed to work within the short moment-arm. This material had a density of 2.3 g/cm^3 , which is almost twice the density of the standard PLA. The nosecone was printed with a 100% infill, which was heavier than required with the intention that excess material may be removed if needed to balance the glider appropriately with the ideal C.G. location. This material costs \$49.99 per 500 grams.
- 4) **Epoxy:** The team was able to source epoxy from 2 different brands (“TotalBoat” and “Aeropoxy”) from the UB AIAA Laboratory, so it added no additional costs to the team. Epoxy pigment was added in the samples of each epoxy, turning them into “epoxy paint”. These samples of paint were tested in 2 scenarios each on separate LW-PLA test pieces. Set A of test pieces had no surface primer applied to it, whereas Set B test pieces were surface primed. As a result, the team discovered that the TotalBoat epoxy would result in a very ‘patchy’ and uneven surface finish. This result varied from the Aeropoxy epoxy, which ended up being cured in a very smooth and even surface finish. Conclusively, the Aeropoxy branded epoxy was used for the final prototype. The test results are shown in the figure below.

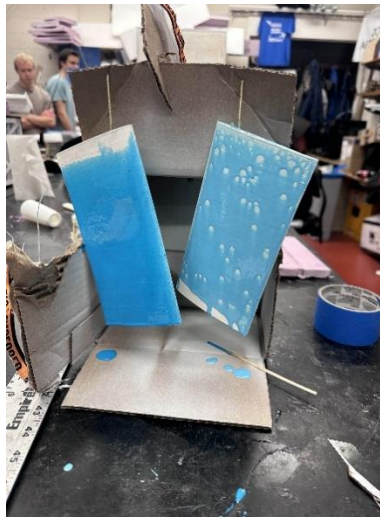


Fig. 12 Epoxy Coating Test with Aeropoxy (Left) and TotalBoat (Right)

B. Component Development

The generation of all parts included consideration of both specific strength and intended purpose. The majority of external parts were hollow structures with 0% infill and comprising of single walled form with minimal reinforcement. Internal structures made of 100% infill were used sparingly lengthwise along the span and length of the fuselage. To properly balance the aircraft's center of gravity, a removeable nose was made of high-density material.

- Wings – the wings sections were developed to be hollow with cross-hatched reinforcement further reduced by a lightening pattern which facilitated an I-beam spar. They parts were super glued together and joined at the wing-box section of the fuselage
- Fuselage – the fuselage section was also developed to be a hollow structure with cross-hatched reinforcement and included channels for high-strength connecting structures. The nose section included a threaded face which allowed the nose to be removed for balancing procedures
- Nose – the nose was designed to be removeable for balancing procedures. Incorporating a threaded connection and composed of high-density material, the nose was sized to make the aircraft nose heavy after initial assembly. Nose material was then drill out to move the center of gravity to the desired point

- Trim-able Horizontal Stabilizer – the tail was designed to include an adjustable deflection to the elevator to facilitate the trim of pitch in flight allowing for adjustment of the glide path
- Internal structures – high-strength internal structures were used to reinforce the low-density external surfaces. Stringers were incorporated to aid the integrity of the fuselage. The wing included a high-strength I-beam made of two C-channels glued together.

VII. 3D Assembly Drawing

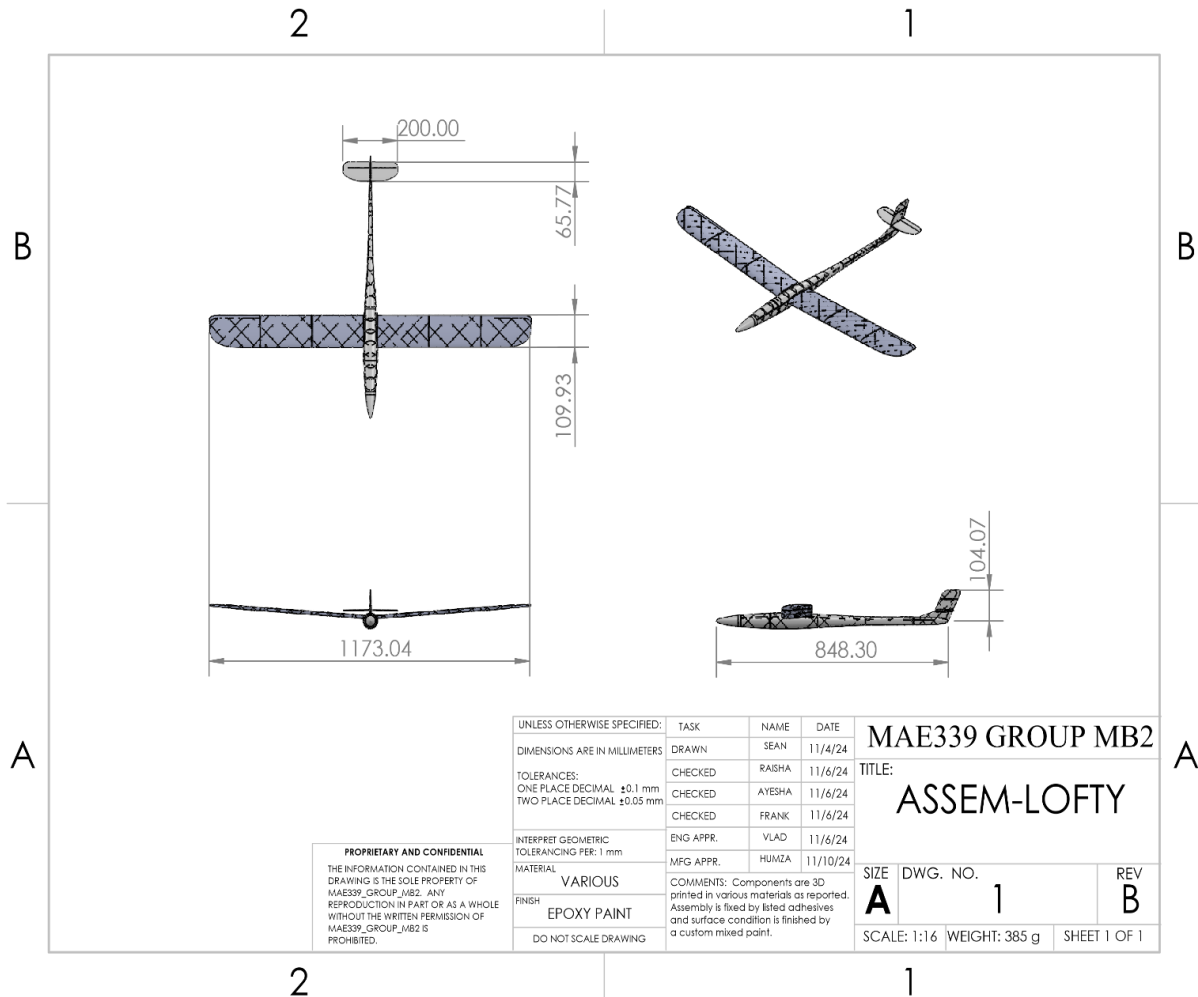


Fig. 13 The Technical 2D drawing of the 3D glider CAD model along with its dimensions

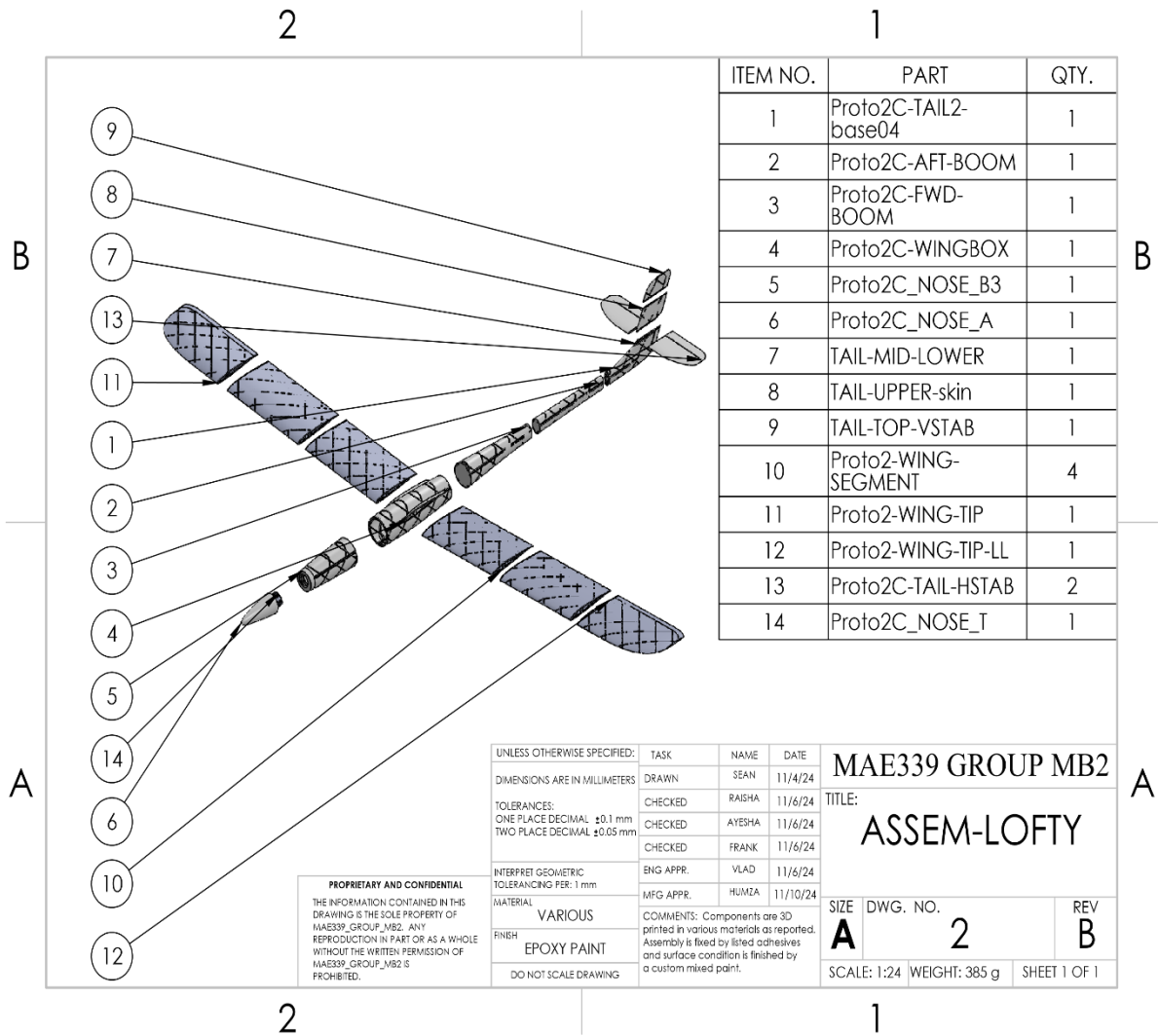


Fig. 14 Isometric View of CAD Model of Glider With its Parts Labeled

VIII. Manufacturing and Assembly

Three different filaments were used to print different components of the aircraft. All components were printed on a Bambu Lab A1 printer. The main fuselage body, along with wings and tails were printed using Lightweight PLA (LW-PLA). LW-PLA offers the benefit of having the lowest print density of all other filaments due to the heat expansion nature of LW-PLA foaming upon heating which would give an overall volume ratio of 220% compared to ordinary PLA. Due to the expansive nature of LW-PLA, structural integrity issues can arise, such as excessive stringing or layer separation (commonly referred to as "grating"), often caused by improper calibration of printing temperature and flow rate. Proper calibration procedure can be found in an online forum by the Bambu community where small test cubes (20mm) were printed at alternating temperatures ranging from 200°C to 280°C. A ruler would then be used to measure all sides of the cube to get an average size. Optimal Temperature was selected with the cube that had the highest average side length. Flow rate calibration then took place at the same temperature to determine which flow produces the best surface finish for the glider parts. Lastly, additional settings were tweaked where the print path must not cross walls at any point which greatly reduced the amount of string shown on parts interior. Results comparison can be seen between the two figures below.



Fig. 15 Print Quality of LW-PLA Before vs. After Calibration

While the fuselage and wings offer a hollow body shell to reduce the most amount of mass while maintaining its structural shape, proper connections between each wing and fuselage piece are needed to hold the aircraft together and add rigidity. These connections are 100% infill PLA strips that run along the channels within the interior of the pieces' interior. For the final filament, Brass filled HTPLA was used to make the nosecone to supply the heavy nosecone weight needed to stabilize the glider.

When all pieces were printed, post-processing took place where all the printed components were given a light sand with 220 grit sandpaper to rid of any stringing that remained on any surface. Once all surfaces were sanded down until the surfaces felt smooth, assembly of the glider followed where superglue was applied on the stringers before slotting them inside the pieces' channels. Once all wings and fuselage components were glued in their respective locations, the horizontal stabilizer was then trimmed to the ideal angle and then weighted on a small scale.

Once the mass of the constructed glider was retrieved, the mass of the glider is then subtracted from the theoretical final mass of the glider. The remaining mass is used as the amount of epoxy resin, hardener, and epoxy pigment paint that is mixed up for the epoxy coating (detail mentioned in Section 5). While the epoxy was mixing, the nosecone of the glider is screwed off and a metal rod is inserted through the glider's fuselage. The excess length of the metal rod protruding was then clamped on a vice which allowed the glider to be overall suspended on all sides for coating.

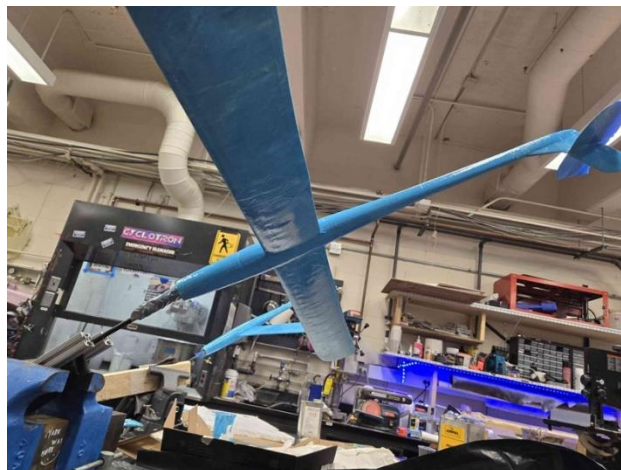


Fig. 16 Glider Suspended via Vice Clamp

Once epoxy was mixed for 5 minutes, two team members dipped paintbrushes into the epoxy mixture and then evenly spread the epoxy mixture across the glider until all surfaces (mainly fuselage and wings) were coated with no gaps of paint. Once the coating was sufficient, the glider was left suspended for at least 48 hours to cure. After the glider was cured, the glider's mass and C.G. were measured again. Depending on the C.G., the weight of the nosecone was designed to be adjusted by drilling holes to remove weight.

IX. Results

In the first prototype of the glider, the team chose to initially use Total Boat Epoxy as the primary resin and hardener to the paint mixture along with an initial tail deflection angle of 12° and a PLA+ nosecone. Some issues that ran during the assembly of the glider lay in the tail section where the low infill percentage caused the small components near the tail to become weak in bending. As a result, some cracks formed when handling the components for assembly and excess superglue was used to reinforce any cracks, increasing the precoat mass. Using Total Boat Epoxy also introduced issues in the coating stage where sections of the wings weren't adhering well to the paint mixture, leaving an uneven coat across the wing which negatively affected aerodynamics. Once the paint cured, the plane was noted to have a significantly tail-heavy center of gravity which forced the nosecone to be taped with lead weights, further harming performance.

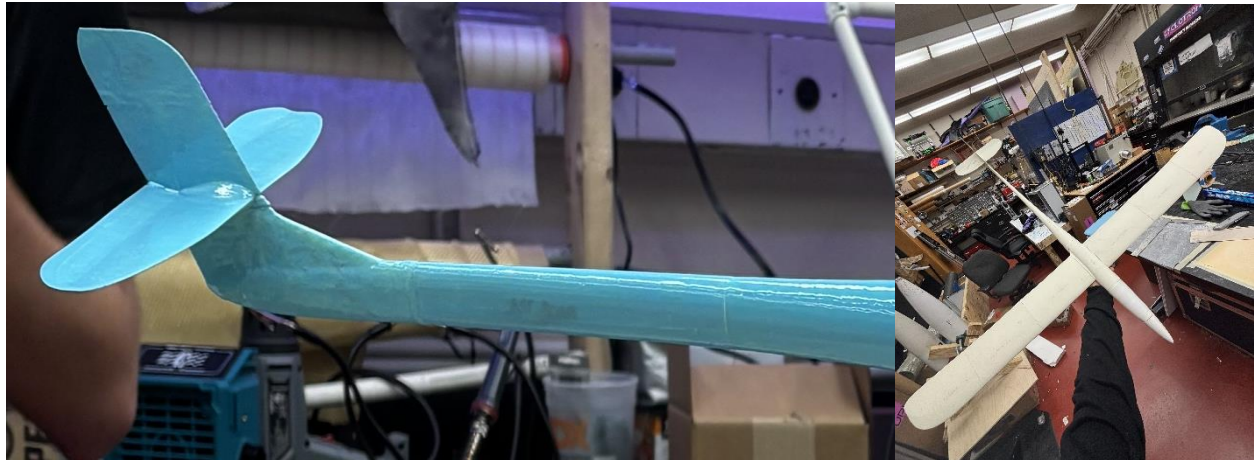


Fig. 17 First Prototype & Tail Deflection Angle

On test flight day, the team headed to the Reservoir State Park in Niagara Falls to measure the total distance the glider can cover. The location was chosen as the tall grass patches in the area would help soften the glider's landing and reduce as much load to the wings and fuselage as possible. Upon the first throw, the glider immediately climbed before stalling and nosediving towards the ground. While the sudden impact force wasn't enough to break the wings, both the tail section and the wing-box fuselage section shattered: equivalent to instant disqualification.



Fig. 18 First Prototype Test Flight and Aftermath

The aftermath of the first test helped us raise some concerns about the design of the glider. Main factors that led to a redesign were:

- Inconsistent finish with selected epoxy.
- Small dimensions of tails were too fragile as 3D prints.
- Nosecone insufficient in stabilizing tail-heavy aircraft.
- The wing-box structure was too weak to support landing impact.

With the gathered data and analysis of the crashed glider, the team reached a conclusion on possible changes that would be introduced in the next 2 gliders printed for competition. One major was redesigning and changing materials of the nosecone to equalize the heavy tail resulted from the coating. Instead of PLA+, a Brass Filled HTPLA filament was used to print the nosecone and utilize the filament's density to leverage more leeway in C.G. control. The nosecone was also elongated in the tip direction by 10 inches for the same reason. Another change implemented into the next prototype was the epoxy change from TotalBoat to Aeropoxy as the team believed the coat job would have improved with a lower density epoxy along with Rust-Oleum primer spray. Additional design and print changes would have been made for the glider but time constraints prevented the team from experimenting further before the day of the competition came.

Print conditions for the next 2 sets of gliders remained the same except for the nosecone piece. The sanding and assembly procedure remained constant as the previous prototype. For the coating stage a primer coat was sprayed onto the glider prior to mixing the epoxy to ensure the bonding of epoxy to the glider's surface. As the epoxy mixture stage came by, Aeropoxy was utilized instead.

X. Conclusion and Improvements

The glider had a relatively good flight performance but failed to land successfully. Here are a few areas that can be improved:

A. Structural weaknesses in certain areas

To address this, additional reinforcements could be incorporated into sections most susceptible to damage. Given more time, employing Finite Element Analysis (FEA) during the design phase could have significantly improved the process. FEA would have allowed us to predict stress distributions and structural vulnerabilities, enabling targeted reinforcement without overdesigning. Although this method requires additional upfront time, it offers the advantage of conserving materials and minimizing iterative redesigns, resulting in a more robust and efficient aircraft structure.

B. Stability Issues

Achieving longitudinal stability remained a significant challenge throughout the project timeline. While a screw-on nose was incorporated to allow for adjustable weight distribution and to shift the center of gravity forward, the aircraft still experienced iterative instability. This issue was exacerbated when reinforcements were added around the tail, frequently making the glider tail-heavy. Despite attempts to adjust the nose weight, the instability persisted, highlighting the need for a more comprehensive approach to balancing structural integrity with aerodynamic performance. Future improvements could include using lighter reinforcement materials, conducting detailed stability analyses, or employing computational tools and aerodynamic simulations to predict and mitigate these issues during the design phase.

C. Long Tail Moment Arm

The tail moment arm was excessively long at over three times to the root chord of the wing away from C.G. Reducing the moment arm length would've allowed for easier balancing, since the tail would've moved in closer despite the tail area increase to maintain the same horizontal tail volume coefficient. Moreover, it would've helped make the tail boom stronger indirectly by having it be shorter.

D. Underutilization of Flight Test Data

While multiple flight tests were conducted, there was limited time and focus dedicated to thoroughly analyzing the results after each test. This oversight meant some valuable insights into performance trends and failure modes were not fully leveraged to guide iterative improvements. Moving forward, dedicating time for systematic post-

flight data analysis and documentation will be crucial. Implementing structured debrief sessions after each test, supported by visual data such as flight recordings and telemetry logs, can help identify performance gaps more effectively and inform targeted design refinements.

Appendix

```
%% Modeling Glider Weight and COM
clear all; close all; clc
```

```
%%%%%%%%%%
% Inputs %
%%%%%%%%%%
```

```
%% Lift coefficient
CL = 0.58;
CLCD = 14.4;
CL_max = 0.9*1.1;
```

```
%% Densities in g/mm^3
rho_wing_1 = 0.5*0.00125;
rho_wing_2 = 0.5*0.00125;
rho_wing_3 = 0.5*0.00125;
rho_wing_spar = 0.00125;
rho_fus_rib = 0.00125;
rho_fus_1 = 0.5*0.00125;
rho_fus_2 = 0.5*0.00125;
rho_fus_3 = 0.5*0.00125;
rho_fus_4 = 0.5*0.00125;
rho_fus_5 = 0.5*0.00125;
rho_HT_1 = 0.5*0.00125;
rho_HT_2 = 0.5*0.00125;
rho_VT = 0.5*0.00125;
```

```
%% Layer thickness in mm
t = 0.44;
```

```
%% Airfoils
Airfoil_wing = "S7012.txt";
Airfoil_tail = "HT12.txt";
```

```
%% Lengths in mm
Root_chord_1 = 110;
Root_chord_2 = 110;
Root_chord_3 = 100;
Tip_chord_3 = 60;
```

```
%% Note: Assume that the start of the wing is from the plane of symmetry.
% That does increases the mass a tiny bit, but the wing-fuselage junction
% fillet isn't included so they somewhat correct each other. Use mm. Use
% half of total span.
wing_length_1 = 480;
wing_length_2 = 80;
wing_length_3 = 30;
```



```

% The layer count for each wing section and spar
wing_layers_1 = 2;
wing_layers_2 = 2;
wing_layers_3 = 2;
wing_spar_layers_1 = 7; % type 0 if no spar in section
wing_spar_layers_2 = 7; % type 0 if no spar in section
wing_spar_layers_3 = 5; % type 0 if no spar in section

%% Fuselage section 1 parameterization in mm. Note: Input the external
% diameters for all fuselage sections
L_1 = 150;
D1 = 50;
fus_layers_1 = 2; % layer count
Nose_weight = 60; % grams

%% Fuselage section 2 parameterization in mm
L_2 = 200;
D2 = D1;
fus_layers_2 = 2; % layer count

% Rib parameterization for section 2
fus_2_rib_layers = 4; % layers here is the depth of the rib in the longitudinal axis
direction
fus_rib_2_width = 5; % width = (Outer diameter - inner diameter) of rib section in mm
fus_rib_loc_2 = [0 100 200]; % rib location from start of section 2 in mm as array

%% Fuselage section 3 parameterization in mm
L_3 = 150;
D3 = 35;
fus_layers_3 = 2; % layer count

% Rib parameterization for section 3
fus_3_rib_layers = 4; % layers here is the depth of the rib in the longitudinal axis
direction
fus_rib_3_width = 4; % width = (Outer diameter - inner diameter) of rib section in mm
fus_rib_loc_3 = [0 75 150]; % rib location from start of section 3 in mm as array

%% Fuselage section 4 parameterization in mm
L_4 = 300;
D4 = 20;
fus_layers_4 = 2; % layer count

% Rib parameterization for section 4
fus_4_rib_layers = 4; % layers here is the depth of the rib in the longitudinal axis
direction
fus_rib_4_width = 3; % width = (Outer diameter - inner diameter) of rib section in mm
fus_rib_loc_4 = [0 140 250]; % rib location from start of section 4 in mm as array

%% Fuselage section 5 parameterization in mm
L_5 = 50;
D5 = D4;
fus_layers_5 = 2; % layer count

%% Horizontal tail parameterization in mm

```

```

% Chord lengths
HT_root_1 = 65;
HT_root_2 = 55;
HT_tip_2 = 30;

% Note: From start of plane of symmetry (use half of total span)
HT_length_1 = 90;
HT_length_2 = 10;

% layer count
HT_layers_1 = 2;
HT_layers_2 = 2;

%% Vertical tail paramaterization in mm

% Chord lengths
VT_root_1 = 70;
VT_root_2 = 60;
VT_tip_2 = 40;

% length of each section
VT_length_1 = 70;
VT_length_2 = 40;

% layer count
VT_layers_1 = 2;
VT_layers_2 = 2;

%%%%%%%%%%%%%%
% Calculations %
%%%%%%%%%%%%%%

% Read airfoil data files
AirfoilDataFile_w = readmatrix(Airfoil_wing);
AirfoilDataFile_t = readmatrix(Airfoil_tail);

% Calculate wing mass & COM
SA_1 = 0;
SA_2 = 0;
SA_3 = 0;
C_wing_1 = zeros(1,100);
C_wing_2 = zeros(1,100);
C_wing_3 = zeros(1,100);
C_spar_1 = zeros(1,100);
C_spar_2 = zeros(1,100);
C_spar_3 = zeros(1,100);
V_spar_1 = zeros(1,100);
V_spar_2 = zeros(1,100);
V_spar_3 = zeros(1,100);
x = AirfoilDataFile_w(:,1);
y = AirfoilDataFile_w(:,2);
[Cx,Cy] = centroid(polyshape(x,y));
y_c4 = y(x>0.2 & x < 0.3);
spar_height_nom = (sum(y_c4(y_c4>0))/length(y_c4(y_c4>0))) -
sum(y_c4(y_c4<0))/length(y_c4(y_c4<0));

```

```

k1 = linspace(Root_chord_1,Root_chord_2,100);
k2 = linspace(Root_chord_2,Root_chord_3,100);
k3 = linspace(Root_chord_3,Tip_chord_3,100);
for ii = 1:length(k1)
    SA_1 = SA_1 + arclength(x*k1(ii),y*k1(ii),'spline')*wing_length_1/100;
    SA_2 = SA_2 + arclength(x*k2(ii),y*k2(ii),'spline')*wing_length_2/100;
    SA_3 = SA_3 + arclength(x*k3(ii),y*k3(ii),'spline')*wing_length_3/100;

    C_wing_1(1,ii) = Cx*k1(ii) + (Root_chord_1 - k1(ii));
    C_wing_2(1,ii) = Cx*k2(ii) + (Root_chord_2 - k2(ii));
    C_wing_3(1,ii) = Cx*k3(ii) + (Root_chord_3 - k3(ii));

    V_spar_1(1,ii) = (wing_length_1/100)*wing_spar_layers_1*t*spar_height_nom*k1(ii);
    V_spar_2(1,ii) = (wing_length_2/100)*wing_spar_layers_2*t*spar_height_nom*k2(ii);
    V_spar_3(1,ii) = (wing_length_3/100)*wing_spar_layers_3*t*spar_height_nom*k3(ii);
    C_spar_1(1,ii) = 0.25*k1(ii) + (Root_chord_1 - k1(ii));
    C_spar_2(1,ii) = 0.25*k2(ii) + (Root_chord_2 - k2(ii));
    C_spar_3(1,ii) = 0.25*k3(ii) + (Root_chord_3 - k3(ii));
end
M_spar_1 = rho_wing_spar*sum(V_spar_1);
M_spar_2 = rho_wing_spar*sum(V_spar_2);
M_spar_3 = rho_wing_spar*sum(V_spar_3);
M_spar = M_spar_1 + M_spar_2 + M_spar_3;
COM_spar = (mean(C_spar_1)*M_spar_1 + mean(C_spar_2)*M_spar_2 +
mean(C_spar_3)*M_spar_3)/(M_spar);
COM_spar = L_1+L_2-Root_chord_1 + COM_spar;
LD_w_1 = rho_wing_1*t*wing_layers_1;
LD_w_2 = rho_wing_2*t*wing_layers_2;
LD_w_3 = rho_wing_3*t*wing_layers_3;
M_wing_1 = SA_1*LD_w_1;
M_wing_2 = SA_2*LD_w_2;
M_wing_3 = SA_3*LD_w_3;
M_wing = (M_wing_1+M_wing_2+M_wing_3)*2;
COM_wing =
1.07*(M_wing_1*mean(C_wing_1)+M_wing_2*mean(C_wing_2)+M_wing_3*mean(C_wing_3))/(0.5*M
_wing);
COM_wing = L_1+L_2-Root_chord_1+COM_wing;
S_wing =
((Root_chord_1)+Root_chord_2)*0.5*wing_length_1+((Root_chord_2)+Root_chord_3)*0.5*wi
ng_length_2+((Tip_chord_3)+Root_chord_3)*0.5*wing_length_3)*2;

% Calculate tail mass and COM
SA_1 = 0;
SA_2 = 0;
SA_3 = 0;
SA_4 = 0;
C_HT_1 = zeros(1,100);
C_HT_2 = zeros(1,100);
C_VT_1 = zeros(1,100);
C_VT_2 = zeros(1,100);
x = AirfoilDataFile_t(:,1);
y = AirfoilDataFile_t(:,2);
[Cx,Cy] = centroid(polyshape(x,y));
k1 = linspace(HT_root_1,HT_root_2,100);
k2 = linspace(HT_root_2,HT_tip_2,100);

```

```

k3 = linspace(VT_root_1,VT_root_2,100);
k4 = linspace(VT_root_2,VT_tip_2,100);
for ii = 1:length(k1)
    SA_1 = SA_1 + arclength(x*k1(ii),y*k1(ii),'spline')*HT_length_1/100;
    SA_2 = SA_2 + arclength(x*k2(ii),y*k2(ii),'spline')*HT_length_2/100;
    SA_3 = SA_3 + arclength(x*k3(ii),y*k3(ii),'spline')*VT_length_1/100;
    SA_4 = SA_4 + arclength(x*k4(ii),y*k4(ii),'spline')*VT_length_2/100;

    C_HT_1(1,ii) = Cx*k1(ii) + (HT_root_1 - k1(ii));
    C_HT_2(1,ii) = Cx*k2(ii) + (HT_root_2 - k2(ii));
    C_VT_1(1,ii) = Cx*k3(ii) + (VT_root_1 - k3(ii));
    C_VT_2(1,ii) = Cx*k4(ii) + (VT_root_2 - k4(ii));
end
LD_HT_1 = rho_HT_1*t*HT_layers_1;
LD_HT_2 = rho_HT_2*t*HT_layers_2;
LD_VT_1 = rho_VT*t*VT_layers_1;
LD_VT_2 = rho_VT*t*VT_layers_2;
M_HT_1 = SA_1*LD_HT_1;
M_HT_2 = SA_2*LD_HT_2;
M_VT_1 = SA_3*LD_VT_1;
M_VT_2 = SA_4*LD_VT_2;
M_HT = (M_HT_1+M_HT_2)*2;
M_VT = (M_VT_1+M_VT_2);
COM_HT = 1*(M_HT_1*mean(C_HT_1)+M_HT_2*mean(C_HT_2))/(0.5*M_HT);
COM_HT = L_1+L_2+L_3+L_4+L_5 - (HT_root_1) + 20 + COM_HT;
COM_VT = 1*(M_VT_1*mean(C_VT_1)+M_VT_2*mean(C_VT_2))/(M_VT);
COM_VT = L_1+L_2+L_3+L_4+L_5 - VT_root_1 + COM_VT;
S_HT =
((HT_root_1)+HT_root_2)*0.5*HT_length_1+((HT_root_2)+HT_tip_2)*0.5*HT_length_2)*2;
S_VT =
((VT_root_1)+VT_root_2)*0.5*VT_length_1+((VT_root_2)+VT_tip_2)*0.5*VT_length_2);

%% Fuselage volume, masses, and COM
Vol_ext_1 = (pi/4)*(1/2)*D1^2*L_1;
t1 = 2*fus_layers_1*t;
Vol_int_1 = (pi/4)*(1/2)*(D1-t1)^2*L_1;
Vol_1 = Vol_ext_1 - Vol_int_1;
COM_f1 = L_1/3;
COM_f1 = L_1 - COM_f1;

Vol_ext_2 = (pi/4)*D2^2*L_2;
t2 = 2*fus_layers_2*t;
Vol_int_2 = (pi/4)*(D2-t2)^2*L_2;
Vol_2 = Vol_ext_2 - Vol_int_2;
COM_f2 = L_2/2;
COM_f2 = L_1 + COM_f2;

Vol_ext_3 = (pi/4)*(1/3)*(D2^2 + D2*D3 + D3^2)*L_3;
t3 = 2*fus_layers_3*t;
Vol_int_3 = (pi/4)*(1/3)*((D2-t3)^2 + (D2-t3)*(D3-t3) + (D3-t3)^2)*L_3;
Vol_3 = Vol_ext_3 - Vol_int_3;
COM_f3_ext =
L_3*((0.5*D2)^2+0.5*(D2*D3)+3*(0.5*D3)^2)/(4*((0.5*D2)^2+0.25*(D2*D3)+0.25*D3^2));
COM_f3_int = L_3*((0.5*(D2-t3))^2+0.5*((D2-t3)*(D3-t3))+3*(0.5*(D3-
t3))^2)/(4*((0.5*(D2-t3))^2+0.25*((D2-t3)*(D3-t3))+0.25*(D3-t3)^2));

```

```

COM_f3 = (Vol_ext_3*COM_f3_ext - COM_f3_int*Vol_int_3)/(Vol_3);
COM_f3 = L_1 + L_2 + COM_f3;

Vol_ext_4 = (pi/4)*(1/3)*(D3^2 + D3*D4 + D4^2)*L_4;
t4 = 2*fus_layers_4*t;
Vol_int_4 = (pi/4)*(1/3)*((D4-t4)^2 + (D4-t4)*(D3-t4) + ((D3-t4)^2))*L_4;
Vol_4 = Vol_ext_4 - Vol_int_4;
COM_f4_ext =
L_4*((((0.5*D3)^2+0.5*(D3*D4)+3*(0.5*D4)^2)/(4*((0.5*D3)^2+0.25*(D3*D4)+0.25*D4^2)));
COM_f4_int = L_4*((((0.5*(D3-t4))^2+0.5*((D3-t4)*(D4-t4))+3*(0.5*(D4-
t4))^2)/(4*((0.5*(D3-t4))^2+0.25*((D3-t4)*(D4-t4))+0.25*(D4-t4)^2)));
COM_f4 = (Vol_ext_4*COM_f4_ext - COM_f4_int*Vol_int_4)/(Vol_4);
COM_f4 = L_1+L_2+L_3+COM_f4;

Vol_ext_5 = (pi/4)*(1/2)*D5^2*L_5;
t5 = 2*fus_layers_5*t;
Vol_int_5 = (pi/4)*(1/2)*(D5-t5)^2*L_5;
Vol_5 = Vol_ext_5 - Vol_int_5;
COM_f5 = L_5/3;
COM_f5 = L_1+L_2+L_3+L_4+COM_f5;

M1 = rho_fus_1*(Vol_1);
M2 = rho_fus_1*(Vol_2);
M3 = rho_fus_1*(Vol_3);
M4 = rho_fus_1*(Vol_4);
M5 = rho_fus_5*(Vol_5);

M_fus = M1 + M2 + M3 + M4 + M5;
M_fus = 79;

COM_fus = (COM_f1*M1 + COM_f2*M2 + COM_f3*M3 + COM_f4*M4 + COM_f5*M5)/M_fus;
COM_fus = 368;

M_rib_2 = rho_fus_rib*0.25*pi*((D2-t2)^2-(D2-t2-fus_rib_2_width)^2)*fus_2_rib_layers;
COM_ribs_2 = sum(L_1+fus_rib_loc_2)/length(fus_rib_loc_2);
M_ribs_2 = length(fus_rib_loc_2)*M_rib_2;

theta_3 = atan((D2-D3)/L_3);
C_rib_3 = 0;
M_rib_3 = zeros(1,length(fus_rib_loc_3));
for ii = 1:length(fus_rib_loc_3)
    D_rib_3 = D2-fus_rib_loc_3(ii)*tan(theta_3);
    M_rib_3(1,ii) = rho_fus_rib*0.25*pi*((D_rib_3-t3)^2-(D_rib_3-t3-
fus_rib_3_width)^2)*fus_3_rib_layers;
    C_rib_3 = C_rib_3 + M_rib_3(1,ii)*(L_1+L_2+fus_rib_loc_3(ii));
end
M_ribs_3 = sum(M_rib_3);
COM_ribs_3 = C_rib_3/sum(M_rib_3);
theta_4 = atan((D3-D4)/L_4);
C_rib_4 = 0;
M_rib_4 = zeros(1,length(fus_rib_loc_4));
for ii = 1:length(fus_rib_loc_4)
    D_rib_4 = D3-fus_rib_loc_4(ii)*tan(theta_4);
    M_rib_4(1,ii) = rho_fus_rib*0.25*pi*((D_rib_4-t4)^2-(D_rib_4-t4-
fus_rib_4_width)^2)*fus_4_rib_layers;

```

```

        C_rib_4 = C_rib_4 + M_rib_4(1,ii)*(L_1+L_2+L_3+fus_rib_loc_4(ii));
    end
    M_ribs_4 = sum(M_rib_4);
    COM_ribs_4 = C_rib_4/sum(M_rib_4);

    M_ribs = M_ribs_2 + M_ribs_3 + M_ribs_4;
    M = Nose_weight + M_ribs + M_wing + M_HT + M_VT + M_spar + M_fus;
    COM = (Nose_weight*COM_f1*(2/3) + M_ribs_4*COM_ribs_4 + M_ribs_3*COM_ribs_3 +
    M_ribs_2*COM_ribs_2 + M_wing*COM_wing + M_HT*COM_HT + M_VT*COM_VT + M_spar*COM_spar +
    M_fus*COM_fus)/M;
    COM_LE = COM - (L_1 + L_2 - Root_chord_1);
    perc_chord_COM = COM_LE/(Root_chord_1);
    M_epoxy = 0.09*M;
    M = M+M_epoxy;

    figure(1)
    title("Relative Weight")
    newcolors =
    ["#538FFF", "#B431E6", "#FF5B58", "#31B800", "#28D2AB", "#FCA207", "#FFDB3D", "#268189", "#2C
    3E50", "#16A085", "#2D1A77"];
    colororder(newcolors)
    piechart([Nose_weight/M, M_ribs/M, M_wing/M, M_HT/M, M_VT/M, M_spar/M, M_fus/M, M_epoxy/M], [
    "Nose Weight", "Fuselage Ribs", "Wing", "Horizontal Stabilizer", "Vertical
    Stabilizer", "Wing Spar", "Fuselage Frame", "Epoxy"]);

    b = (wing_length_1 + wing_length_2 + wing_length_3)*2;
    AR = b^2/S_wing;
    L_VT = (COM_VT - (L_1 + L_2 - 0.75*Root_chord_1));
    L_HT = (COM_HT - (L_1 + L_2 - 0.75*Root_chord_1));
    VTVC = L_VT*S_VT/(b*S_wing);
    HTVC = L_HT*S_HT/(Root_chord_1*S_wing);
    Vel = sqrt((2*(M/1000)*9.81)/(1.225*(S_wing/10^6)*sqrt(CL^2 + (CL/CLCD)^2)));
    Vstall = sqrt((2*(M/1000)*9.81)/(1.225*CL_max*(S_wing/10^6)));
    fprintf(" \nWingspan [in]: %f \nWing Area [m^2]: %f \nAspect Ratio: %f \n" + ...
    "Wing Loading [Kg/m^2]: %f \nFuselage Length [in]: %f \nCG [Perc. Chord]: %f
    \n"+...
    "Mass [grams]: %f \nHorizontail Tail Volume: %f \nVertical Tail Volume: %f\nStall
    speed [m/s]: %f \nVelocity [m/s]: %f \n\n", b*39.37/1000, S_wing*10^-
    6, AR, (M/1000)/(S_wing*10^-6), +...
    (L_1+L_2+L_3+L_4+L_5)*39.37/1000, perc_chord_COM*100, M, HTVC, VTVC, Vstall, Vel);

```

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